

Kunal Maheshwari

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SUMMARY

OPT-EAD holder with full-time US work authorization, bringing over 4 years of professional experience as a Software Developer and Data Engineer. Expertise in building scalable microservices and distributed data pipelines using Java, Python, Spark, SQL, AWS, and Azure, complemented by CI/CD automation and graduate-level work on large-scale data platforms and privacy-preserving AI. Proven ability in architecture design, API design, and full-stack development, with a strong command of Go, Rust, C/C++, and modern JavaScript frameworks.

EDUCATION

Central Michigan University | Mount Pleasant, MI Aug 2024 - Dec 2025
Master of Science | Computer Science

Lovely Professional University | Punjab, India Aug 2016 - May 2020
Bachelor of Technology | Computer Science and Engineering

EXPERIENCE

Software Developer | NTT DATA | Bengaluru, India (Remote) August 2022 - August 2024

- Architected and optimized high-performance REST and SOAP microservices using Java 8/11 and Spring Boot, handling over 50,000 daily API requests with 99.99 percent SLA uptime, demonstrating strong API Design capabilities.
- Engineered and automated robust CI/CD pipelines with Jenkins, Maven, and Azure DevOps, facilitating continuous delivery across Azure Cloud and Oracle WebLogic environments.
- Modernized legacy systems by migrating to Log4j 2.x, enhancing monitoring and error handling to reduce issue resolution time by 20 percent.
- Led a significant EDX platform upgrade, streamlining application architecture and dependency management to achieve 50 percent faster delivery cycles.
- Collaborated within Agile and Scrum teams to deliver FHIR-compliant healthcare solutions, utilizing Git, SVN, and comprehensive JUnit testing.

Senior Analyst | Capgemini | Mumbai, India (Remote) October 2021 - August 2022

- Engineered enterprise applications and RESTful web services using Java and EJB for automotive clients, focusing on Software Engineering principles.
- Improved system performance by 90 percent through SQL query optimization, database tuning, and modernization of legacy Java code, showcasing Benchmarking expertise.
- Implemented CI/CD pipelines leveraging Jenkins and GitLab, integrating automated testing and quality gates for enhanced code integrity.
- Mentored junior developers through detailed code reviews, debugging sessions, and the promotion of clean coding practices.

Analyst | Capgemini | Mumbai, India (Remote) July 2020 - October 2021

- Maintained high-availability enterprise systems with over 99 percent uptime through proactive monitoring and performance optimization.
- Enhanced SQL data pipelines and database operations, improving query performance by 10 percent and strengthening data integrity.
- Implemented comprehensive unit testing using JUnit4, achieving 99 percent code coverage and adhering to Test Driven Development principles.
- Supported release planning and sprint execution using JIRA within Agile development teams.

PROJECTS

NYU Taxi Data Analysis Platform (Spark, Python, AWS, Terraform) January 2024 - May 2025

- Developed a distributed Spark-based data pipeline to process and analyze over 8 GB of NYC taxi trip data in near real-time.
- Built an interactive Flask web application for data exploration with dynamic filtering and visualization, showcasing Fullstack development.
- Containerized ingestion, processing, and analytics components using Docker for reproducible deployments.
- Optimized Spark memory usage and partitioning strategies, reducing overall job runtime by 40 percent.
- Automated cloud infrastructure provisioning using Terraform across AWS services including S3, RDS, ECS, Lambda, and QuickSight, demonstrating Infrastructure automation.

Privacy-Preserving AI Email Client (Capstone Project - T5, LoRA, Python) January 2025 - December 2025

- Engineered a fine-tuning pipeline for T5-based transformer models to perform email summarization and classification, utilizing Hugging Face libraries.
- Applied parameter-efficient fine-tuning using LoRA for low-resource training under privacy constraints, contributing to Privacy-preserving LLM-based applications.
- Implemented model evaluation using ROUGE metrics and integrated quantization for secure local deployment.

SKILLS

- Programming Languages: Proficient in Java, Python, Go, Rust, C/C++, and SQL for robust Software Engineering and Architecture Design.
- Frameworks and Libraries: Extensive experience with Spring Boot, Spring Framework, Apache Spark, PySpark, Hugging Face, T5, LoRA, and JUnit for building scalable applications.

- Web Services & APIs: Expertise in REST API and SOAP API development, with a strong foundation in API Design and Stripe/Paypal integrations.
- Cloud Platforms: Skilled in AWS (EC2, RDS, S3, Lambda), Azure (Functions, DevOps, ServiceBus), and Google Cloud Platform, supporting enterprise-level deployments.
- Databases & Data Tools: Proficient with PostgreSQL, Oracle Database, IBM DB2, RDS, and advanced data pipeline techniques (ETL, Data Warehousing) using tools like DBeaver and Tableau.
- Development Tools & Concepts: Experienced in Docker, Kubernetes, Jenkins, Git, SVN, Maven, JIRA, Confluence, and Terraform for CI/CD, Prototyping, and Infrastructure automation.
- Data Analysis & Visualization: Adept at Data Analytics and Visualization using Tableau and QuickSight, driving insights and supporting decision-making.
- Methodologies & Practices: Strong understanding of Agile, Scrum, Test Driven Development, Object Oriented Programming, Microservices, and Large-scale data platforms.